



Games > Actions

## Games

# Actions

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## Input interface for ARC-AGI-3 games

All games implement a standardized action interface with seven core actions:

| Action  | Description                                                                                   |
|---------|-----------------------------------------------------------------------------------------------|
| RESET   | Initialize or restarts the game/level state                                                   |
| ACTION1 | Simple action - varies by game (semantically mapped to up)                                    |
| ACTION2 | Simple action - varies by game (semantically mapped to down)                                  |
| ACTION3 | Simple action - varies by game (semantically mapped to left)                                  |
| ACTION4 | Simple action - varies by game (semantically mapped to right)                                 |
| ACTION5 | Simple action - varies by game (e.g., interact, select, rotate, attach/detach, execute, etc.) |
| ACTION6 | Complex action requiring x,y coordinates (0-63 range)                                         |
| ACTION7 | Simple action - Undo (e.g., interact, select)                                                 |

## Human Player Keybindings

When playing games manually in the ARC-AGI-3 UI, you can use these keyboard shortcuts instead of clicking action buttons:

| Control Scheme | ACTION1 | ACTION2 | ACTION3 | ACTION4 |
|----------------|---------|---------|---------|---------|
| WASD + Space   | W       | S       | A       | D       |
| Arrow Keys + F | ↑       | ↓       | ←       | →       |



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## Game over state

When a game reaches a game-over state, the only valid action is `RESET`. Sending any other action (e.g., `ACTION1` through `ACTION7`) to a game in this state returns a `400 Bad Request` error.

If you encounter a `400` error during gameplay, check whether the game has ended and issue a `RESET` to start a new game.

! A `400` error indicates the game is in a game-over state. A `500` error, by contrast, indicates a server-side issue.

## Available actions

Each game explicitly defines the set of available actions that can be used within that game. This approach ensures clarity for both human and AI participants by making it clear which actions are permitted, thereby reducing confusion. In the human-facing UI, available actions are visually highlighted or dismissed to provide the same affordance.

For each action taken, the metadata of the returned frame will indicate which actions are available. Agents may use this information to narrow the action space and develop effective strategies for completing the game.

Note: Action 6 does not provide explicit X/Y coordinates for active areas. If Action 6 is available, only its availability will be indicated, without specifying which coordinates are active.

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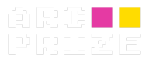
Yes

No

Suggest edits

< Available Games

Methodology >



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